Appendix

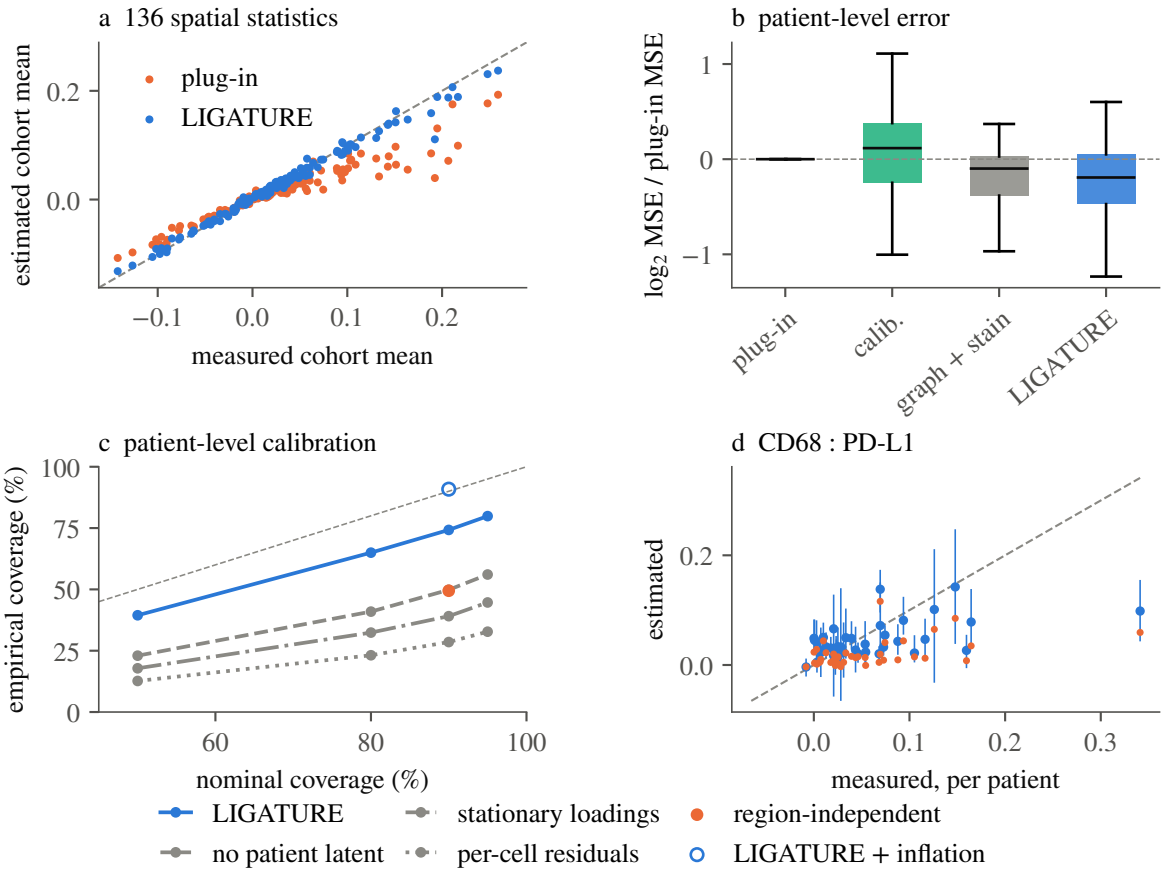


Figure 3: **a**, Cohort-mean estimate against measured value for the 136 statistics. **b**, Reference-run patient-level MSE relative to the plug-in, per statistic (boxes: quartiles; whiskers: 1.5 IQR). The direct graph uses stain inputs and is the matched Table 1 comparator; the weaker direct ridge is omitted. **c**, Empirical against nominal coverage of patient-level intervals, averaged over statistics, for LIGATURE, its ablations without patient latent or with stationary loadings, and calibrated per-cell residuals sampled independently. **d**, CD68–PD-L1 co-localization with reference-run raw 90% intervals; the high-measured-value patient is missed, illustrating undercoverage.

Identification computations. The endpoints are max-cut-type semidefinite programs, which we solve by Riemannian ascent on $R = VV^\top$ [28] (unit rows, rank $k > \sqrt{2n}$), certify with the dual bound $\sum_i y_i + n \max(0, -\lambda_{\min}(\mathrm{diag}(y) - \Omega))$, $y_i = (\Omega V V^\top)_{ii}$, and round into explicit rank-one Gaussian witnesses $Y_i = \mu_i + \sigma_i s_i Z$, $s = \mathrm{sign}(Vg)$ [29]. A procedure that specifies only per-cell predictive laws (heteroscedastic heads, per-cell conformal intervals, ensembles or multiple imputation sampled independently per cell) determines a spatial statistic only up to this interval; we compute it on real tissue in Section 6. Hyperplane rounding gives explicit rank-one Gaussian couplings with the prescribed per-cell laws that reach a median 90% (minimum 80%) of the upper endpoint, and the dual bound is within 0.33% of every computed upper endpoint and 4.9% (median 0.26%) of every lower endpoint; certified outer bounds widen patient-level intervals by at most 0.29%. The restricted interval uses stationary, nonnegative correlations decaying with distance: mixtures of exponential kernels with ranges 5–160 μm plus a nugget, for which the endpoints are closed form.

Patient-level uncertainty.

Proposition 5 (Patient-level uncertainty)**.** *For a patient with region statistics $Q_1, \ldots, Q_m$,* $\mathrm{Var}(\bar{Q} \mid H) = m^{-2} \sum_c \mathrm{Var}(Q_c \mid H) + m^{-2} \sum_{c \neq c'} \mathrm{Cov}(Q_c, Q_{c'} \mid H)$. *A generator that samples regions independently sets the second term to zero and, with equal region variances, understates the variance by the design effect* $1 + (m-1)\bar{\rho}$

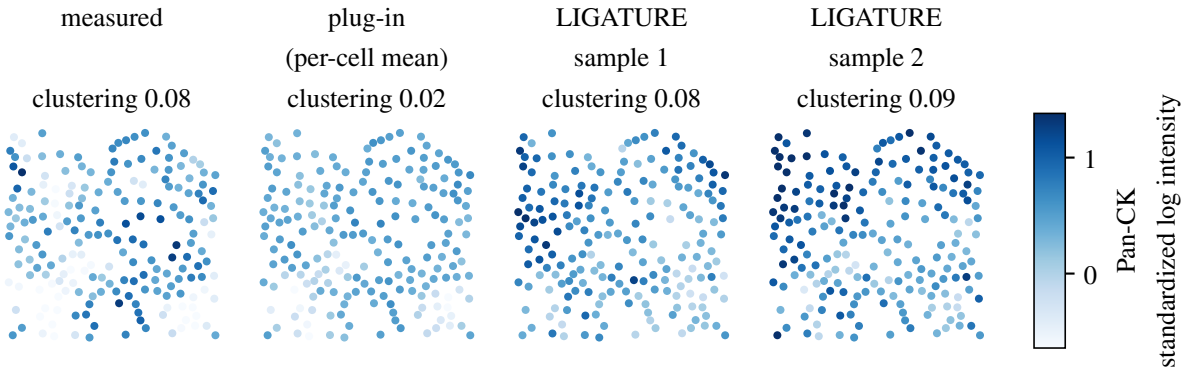


Figure 4: An illustrative held-out region (patient CRC01, 196 cells), selected automatically as the largest of the 15 test regions whose measured Pan-CK clustering is closest to the fold median. The measured near-minus-far contrast is 0.08, the plug-in gives 0.02, and two model draws give 0.08 and 0.09. These draws illustrate the representation used for statistical inference, with the frozen stain as their mean.

for average cross-region correlation $\bar{\rho}$.

Statistic-averaged split conformal.

Proposition 6 (Statistic-averaged split conformal)**.** *Let patients* $1, \ldots, n+1$ *be exchangeable, with normalized scores* $r_{iq} = |Q_{iq} - \hat{\mu}_{iq}|/\hat{s}_{iq}$ *for statistics* $q = 1, \ldots, J$*, where* $\hat{\mu}, \hat{s}$ *are the generator's sample mean and standard deviation, computed by a model that did not see these patients. Let* $g_i(t) = J^{-1} \sum_q \mathbf{1}\{r_{iq} \le t\}$ *and* $\hat{\tau} = \inf\{t : (n+1)^{-1} \sum_{i \le n} g_i(t) \ge 1 - \alpha\}$*. Then the intervals* $\hat{\mu}_{n+1,q} \pm \hat{\tau}\hat{s}_{n+1,q}$ *satisfy* $\mathbb{E}[g_{n+1}(\hat{\tau})] \ge 1 - \alpha$*: the expected fraction of a new patient's statistics that are covered is at least* $1 - \alpha$*.*

Proof. Let $A(t) = (n+1)^{-1} \sum_{i \le n+1} g_i(t)$ and $\tau^* = \inf\{t : A(t) \ge 1-\alpha\}$, a symmetric function of the $n+1$ patients. Since $g_{n+1} \ge 0$, $(n+1)^{-1} \sum_{i \le n} g_i(t) \le A(t)$ for all t, so $\{t : (n+1)^{-1} \sum_{i \le n} g_i(t) \ge 1 - \alpha\} \subseteq \{t : A(t) \ge 1-\alpha\}$ and $\hat{\tau} \ge \tau^*$; as g_{n+1} is nondecreasing, $\mathbb{E}[g_{n+1}(\hat{\tau})] \ge \mathbb{E}[g_{n+1}(\tau^*)] = \mathbb{E}[(n+1)^{-1} \sum_{i \le n+1} g_i(\tau^*)] \ge 1 - \alpha$ by exchangeability. □ This is conformal risk control [22] with the miscovered-fraction loss, bounded by one. The guarantee holds for dependent statistics within a patient and needs $1 - \alpha \le n/(n+1)$; with $n = 8$ validation patients we use $\alpha = 0.15$.

Table 4: Earlier ablations of the main run (same 40 held-out patients and 136 statistics), including independently sampled per-cell residuals (multiple imputation). Raw coverage is of nominal 90% intervals before inflation.

Variant	MSE all	MSE same	bias	CRPS	coverage
LIGATURE	0.88	0.60	−6%	0.027	74.2%
without variance calibration	0.86	0.59	−8%	0.029	54%
cell-level score only	2.40	2.67	−23%	0.045	49%
without spatial residual fields	1.03	0.66	0%	0.029	79%
without patient latent	0.86	0.59	−5%	0.029	49.8%
stationary loadings	0.92	0.67	−15%	0.032	39%
marginal only (multiple imputation)	1.01	1.00	−34%	0.036	28%

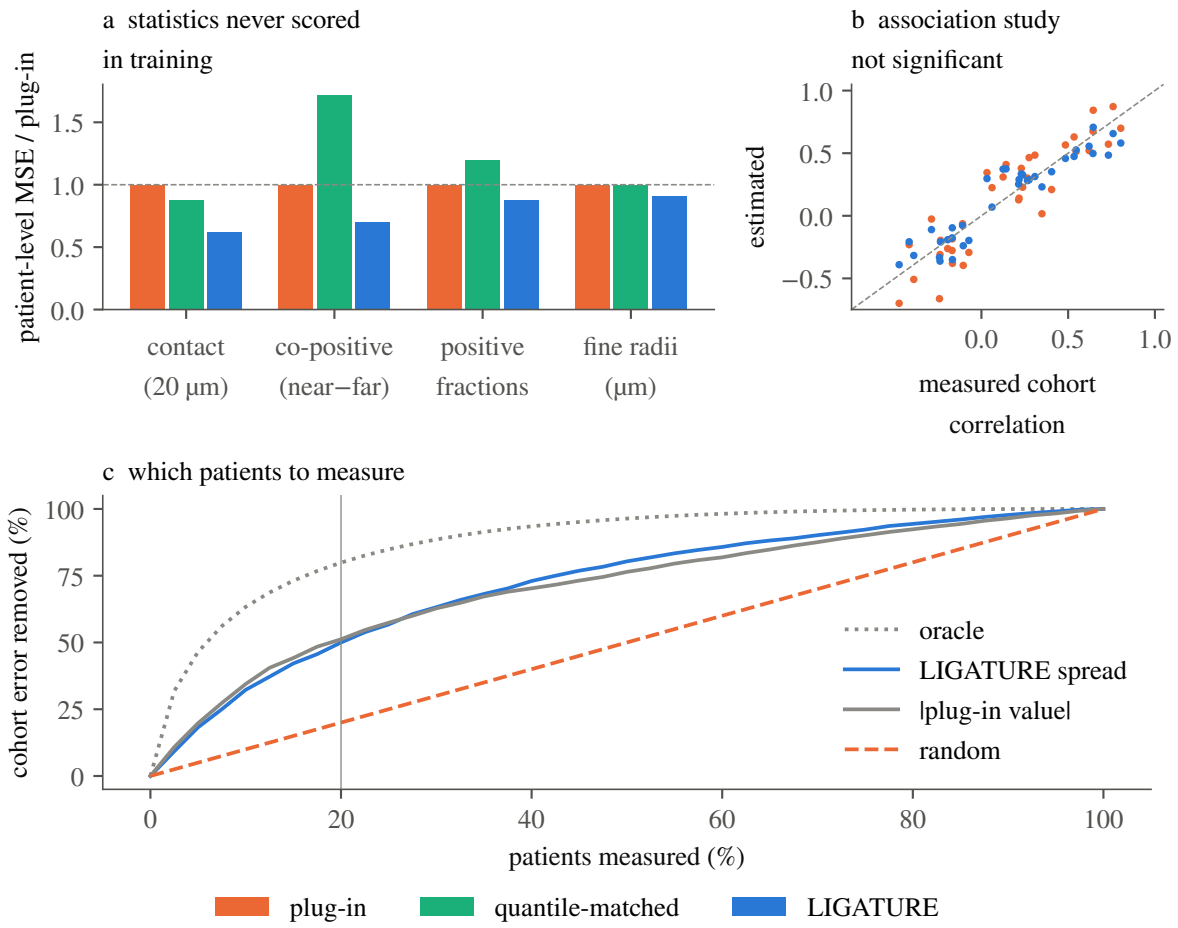


Figure 5: **a**, Patient-level MSE for statistic families never scored in training: gated contact fractions within 20 μm (256 ordered pairs), near-minus-far co-positivity, positive-cell fractions, and products at 0–12.5 versus 12.5–25 μm. **b**, Spearman correlation across 40 patients between pairs of nine predefined spatial biomarkers. The reduction in correlation-estimation error is not statistically significant (90% bootstrap interval includes zero). **c**, Fraction of cohort squared error removed by measuring patients in order of LIGATURE's predictive spread, of the magnitude of the plug-in value (a mean-only control), at random, or by an oracle, averaged over the 136 statistics.

Table 5: Systematic ablations: each variant is the reference configuration with one change, five patient-held-out rotations (ORION 40 patients, lung 19). MSE is patient-level MSE relative to the plug-in over all 136 statistics and the 16 clustering statistics; ρ mean Spearman correlation; bias median relative bias; coverage of nominal 90% intervals raw and after inflation. *: some metric outside the range of the reference seeds. ORION's seeds are current-code runs of the reference configuration; the reference run of Table 1 is a separate, earlier run. Per-patient Wilcoxon tests against the reference are in `research/ABLATIONS.md`.

Cohort	Variant	MSE all	MSE same	ρ	bias	CRPS	cov. raw / infl.
ORION	current-code seed 1	0.94	0.63	0.52	-6%	0.0282	72 / 90
	current-code seed 2	0.90	0.60	0.52	-8%	0.0272	74 / 89
	current-code seed 3	0.91	0.60	0.52	-5%	0.0280	73 / 90
ORION	cell-level score only*	1.33	1.22	0.45	-15%	0.0360	52 / 89
	no cell-level score	0.92	0.62	0.53	-5%	0.0280	73 / 90
	bin contrasts not scored*	1.29	0.84	0.43	+8%	0.0353	57 / 90
	no patient-level score*	0.91	0.59	0.54	-6%	0.0274	73 / 91
	no region-level score*	0.94	0.58	0.53	-7%	0.0277	72 / 90
ORION	no spatial fields*	1.00	0.62	0.50	+2%	0.0282	82 / 89
	no region or patient latent*	1.02	0.68	0.46	+1%	0.0327	45 / 88
	no patient latent	0.90	0.61	0.52	-4%	0.0301	49 / 88
	no region latent	0.94	0.61	0.50	+0%	0.0281	78 / 89
	field rank 1	0.92	0.63	0.50	-5%	0.0277	79 / 89
	field rank 8*	0.90	0.59	0.53	-5%	0.0274	73 / 91
	one field scale (20 um)	0.92	0.63	0.52	-6%	0.0277	78 / 90
	coarse scales only (40, 80 um)	0.93	0.61	0.50	-5%	0.0279	76 / 90
	fine scales only (10, 20 um)	0.91	0.60	0.52	-6%	0.0274	79 / 92
	stationary loadings*	0.92	0.67	0.51	-15%	0.0320	39 / 88
ORION	width 64*	0.86	0.59	0.54	-1%	0.0274	75 / 91
	no dropout*	0.89	0.59	0.53	-5%	0.0273	78 / 92
	no morphology descriptors	0.93	0.61	0.53	-5%	0.0278	75 / 91
	residual network does not see the stain*	0.92	0.58	0.53	-4%	0.0276	74 / 90
	1 transformer layer*	0.88	0.62	0.51	-7%	0.0276	74 / 90
ORION	5 epochs*	0.89	0.58	0.52	-3%	0.0273	76 / 91
	4 samples per region in the score	0.90	0.60	0.52	-8%	0.0278	72 / 90
	12 training patients	0.90	0.60	0.52	-7%	0.0277	73 / 91
	6 training patients*	0.90	0.63	0.50	-8%	0.0283	73 / 90
lung	reference	0.84	0.48	0.48	-23%	0.0182	64 / 89
	reference, seed 2	0.86	0.51	0.48	-27%	0.0188	63 / 89
lung	cell-level score only*	2.27	2.06	0.34	-49%	0.0286	56 / 89
	no patient-level score*	0.83	0.51	0.49	-30%	0.0188	60 / 89
	no region-level score*	0.86	0.51	0.46	-22%	0.0186	65 / 89
lung	no spatial fields*	0.94	0.58	0.47	-24%	0.0187	81 / 90
	no region or patient latent*	0.86	0.50	0.48	-20%	0.0192	47 / 89
	stationary loadings*	0.94	0.64	0.41	-43%	0.0222	44 / 89
lung	residual network does not see the stain*	0.86	0.52	0.47	-25%	0.0189	61 / 89
lung	6 training patients*	0.85	0.53	0.49	-25%	0.0187	62 / 89

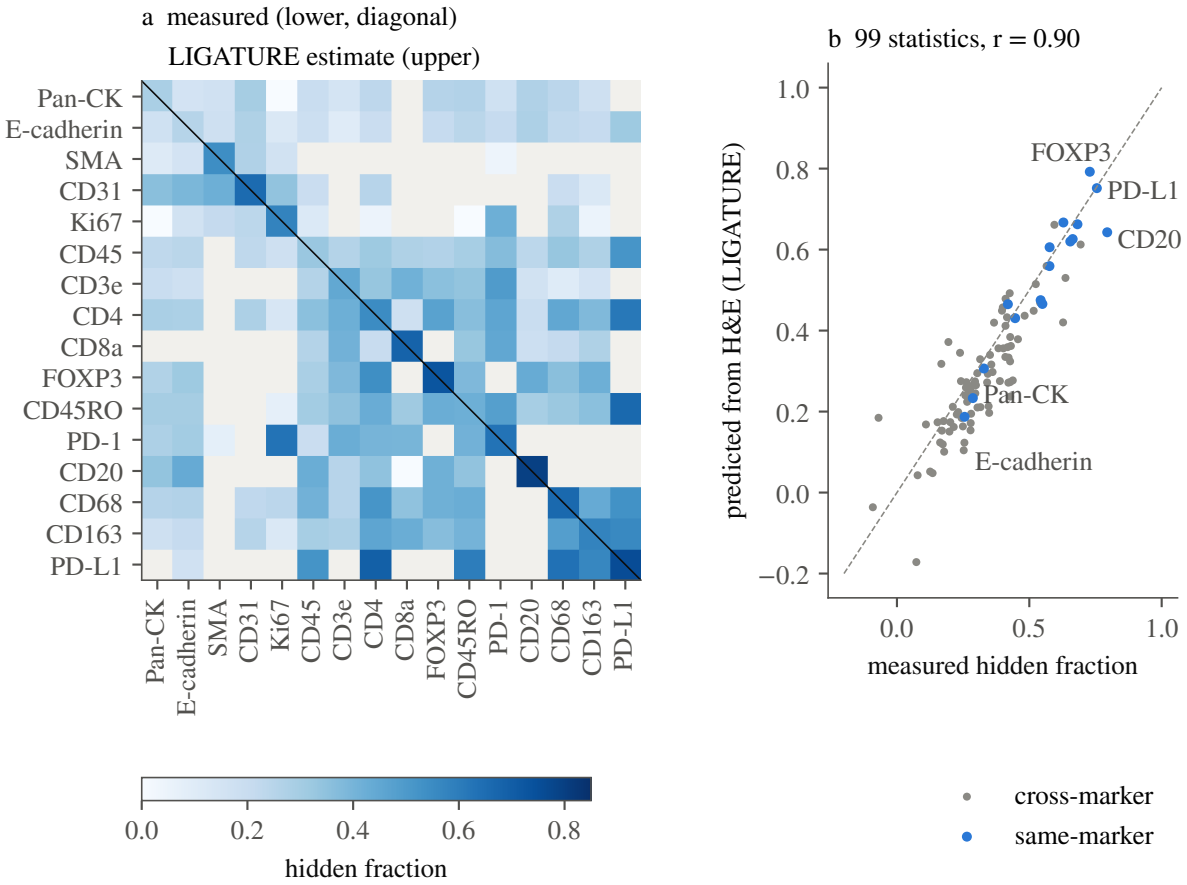


Figure 6: **a**, Fraction of each cohort-mean spatial statistic missing from the per-cell virtual stain, measured (lower triangle and diagonal) and estimated by LIGATURE from H&E only on held-out patients (upper triangle); gray: |measured mean| ≤ 0.02. **b**, Estimated against measured hidden fraction.

A1 Proofs

Proposition 1. Write $Y = \mu + \varepsilon$ with $\mathbb{E}[\varepsilon \mid H] = 0$. Then $Y_{ia}Y_{jb} = \mu_{ia}\mu_{jb} + \mu_{ia}\varepsilon_{jb} + \varepsilon_{ia}\mu_{jb} + \varepsilon_{ia}\varepsilon_{jb}$, and $\mathbb{E}[\cdot \mid H]$ of the two middle terms vanishes, so $\mathbb{E}[Q(Y) \mid H] = Q(\mu) + |E_w|^{-1} \sum_{i \neq j} w(d_{ij}) \frac{1}{2}(\Sigma_{ia,jb} + \Sigma_{ib,ja})$. Subtracting $Q(f)$ gives the claim. Nothing is assumed about f beyond measurability with respect to H.

Proposition 2, cross-marker case. With bivariate per-cell laws $\mathcal{N}(\mu_i, \Sigma_i)$, $\Sigma_i = \left(\begin{smallmatrix} \sigma_{ia}^2 & \rho_i \sigma_{ia}\sigma_{ib} \\ \rho_i \sigma_{ia}\sigma_{ib} & \sigma_{ib}^2 \end{smallmatrix}\right)$, any joint law of the $2n$ variables with these marginals has a correlation matrix in $\mathcal{F} = \{R \in \mathcal{E}_{2n} : R_{(i,a),(i,b)} = \rho_i\}$, a convex set, and every $R \in \mathcal{F}$ is attained by a Gaussian law with these marginals. $\mathbb{E}[Q \mid H] - Q(\mu)$ is linear in R, so its range is an interval with endpoints given by two semidefinite programs over $\mathcal{F}$. Non-Gaussian marginals restrict the attainable correlations further (Fréchet–Hoeffding bounds), so the Gaussian interval contains the interval for any marginals with the same means and variances whose copulas are unrestricted.

Proposition 3. (i) A bin-constant weight $w = \sum_k c_k \mathbf{1}_{B_k}$ gives $Q_{w,ab} = \sum_k c_k |E_k| \, G_{k,ab}/|E_w|$, which is linear in S for fixed geometry. (ii) For each j, $\mathbb{E}\,\mathrm{CRPS}(F, \theta_j(S))$ is uniquely minimized over laws with finite mean at the true conditional law of $\theta_j(S)$ (strict propriety); a sum of such terms is minimized exactly when every term is, i.e. by any joint predictive law with the correct marginals along every θ_j. The cell term identifies each per-cell marginal in the same way.

Proposition 4. (i) The hypothesis requires f to reproduce the scored cell means, since a zero-mean residual cannot shift them; for a stain with region-mean error the hypothesis fails and a fit is a misspecified projection. Under the hypothesis, by Proposition 3(ii) a minimizer's law of each scored coordinate equals the true conditional law, in particular its mean. Under the model $Y = f + R$ with $\mathbb{E}[R \mid H] = 0$, the cross terms of $G_{k,ab}(f + R)$ vanish in expectation, so $\mathbb{E}[G_{k,ab}(Y) \mid H] = G_{k,ab}(f) + \Gamma_{k,ab}(\Sigma)$; equating means gives the identity. A bin-constant bilinear statistic is a linear combination of the $G_{k,ab}$ (Proposition 3(i)), so its mean is also correct. (ii) Cell means do not depend on Σ, the mean of $G_{k,ab}$ depends on it only through $\Gamma_{k,ab}$, and contrasts and patient averages are linear in these. A perturbation Δ with $\Gamma_{k,ab}(\Delta) = 0$ for all k, a, b leaves all these means unchanged; an example is $\Delta_{ia,ib} = \Delta_{ib,ia} = \epsilon$ for one cell i and $a \neq b$, which enters no bin because bins exclude $i = j$, and keeps $\Sigma + \Delta$ positive definite for small ϵ when Σ is. It changes $\mathbb{E}[Y_{ib} \mid Y_{ia}, H] = f_{ib} + \Sigma_{ib,ia}\Sigma_{ia,ia}^{-1}(Y_{ia} - f_{ia})$. The cell-level score constrains only the per-cell marginal variances $\Sigma_{ia,ia}$. Such perturbations still change the spread of scored statistics, through fourth moments, so they are weakly constrained by the full scores; they are not constrained by the means that determine point estimates.

Proposition 7. (i)

$$\partial_{\theta_k}\big[\mathrm{tr}(\Sigma_\theta^{-1}\Sigma^*) + \log\det\Sigma_\theta\big] = -\mathrm{tr}(\Sigma_\theta^{-1}K_k\Sigma_\theta^{-1}\Sigma^*) + \mathrm{tr}(\Sigma_\theta^{-1}K_k) = -\mathrm{tr}\big(\Sigma_\theta^{-1}K_k\Sigma_\theta^{-1}(\Sigma^* - \Sigma_\theta)\big),$$

which vanishes at an interior KL projection; and with a frozen mean equal to the true conditional mean, $\mathbb{E}^*[Q_W(Y)] - \mathbb{E}_\theta[Q_W(Y)] = \mathrm{tr}\big(W(\Sigma^* - \Sigma_\theta)\big)$. These are the likelihood and moment equations of classical variance-component estimation; the statistic score chooses the moments to be the analyzed statistics. (ii) With $\Sigma^* = I + \rho A$ (A the bin-1 adjacency) and $\Sigma = s^2 I + c^2 \mathbf{1}\mathbf{1}^\top$, whose eigenvalues are $\lambda_1 = s^2 + nc^2$ on $\mathbf{1}$ and $\lambda_0 = s^2$ on its complement, the KL projection minimizes $\mathrm{tr}(\Sigma^{-1}\Sigma^*) + \log\det\Sigma = a/\lambda_1 + (n-a)/\lambda_0 + \log\lambda_1 + (n-1)\log\lambda_0$ with $a = \mathbf{1}^\top\Sigma^*\mathbf{1}/n = 1 + 2\rho N_1/n$. The minimizer is $\lambda_1 = a$, $\lambda_0 = (n-a)/(n-1)$, so for $\rho \geq 0$ $c^2 = (\lambda_1 - \lambda_0)/n = 2\rho N_1/(n(n-1)) = \rho N_1/\binom{n}{2}$ (for $\rho < 0$ the constrained optimum is $c^2 = 0$) and $s^2 + c^2 = \lambda_0 + (\lambda_1 - \lambda_0)/n = 1$. The bin-1 moment has mean $\Gamma_1 = c^2$ under the class and ρ under the truth, so matching it gives $c^2 = \rho$. We confirmed the closed form by numerical KL minimization on random neighbor graphs.

Proposition 5. $\mathrm{Var}(\bar{Q}) = m^{-2}\sum_{c,c'}\mathrm{Cov}(Q_c, Q_{c'})$. With equal variances v and average correlation $\bar{\rho}$ over ordered pairs $c \neq c'$, this is $m^{-1}v(1 + (m-1)\bar{\rho})$, versus $m^{-1}v$ under independence.

Proposition 6. Given in the main text. Ties are handled by the infimum over a right-continuous step function; the bound does not require independence across statistics within a patient.

A2 Implementation details

Crop selection and windows. The supplied crop release (`https://huggingface.co/datasets/csmp-hub/orion-crc`) contains 3,337 segmentation-selected regions, not all cells of the original whole slides. The

Table 6: Reference patient roles: four-run aggregate versus the direct reseed mean on 40 ORION patients. Each metric averages the per-run patient-MSE ratio; this is not prediction ensembling. LIGATURE uses the reference plus three current-code reseeds; direct prediction uses three seeds. Brackets are 95% patient-bootstrap intervals conditional on fitted models (2,000 replicates); the final row is a paired difference, with positive values favoring direct prediction.

Estimator (runs)	MSE all	MSE cross	MSE same	range: all; same
Global rescale (1)	0.89 [0.83, 1.02]	0.90 [0.83, 1.05]	0.83 [0.76, 0.89]	–
Direct graph (stain inputs), reseed mean (3)	0.88 [0.85, 0.95]	0.92 [0.88, 0.98]	0.64 [0.59, 0.68]	0.867–0.892; 0.629–0.655
LIGATURE four-run aggregate (4)	0.91 [0.87, 1.02]	0.95 [0.92, 1.06]	0.61 [0.55, 0.75]	0.883–0.945; 0.596–0.630
LIGATURE − direct	+0.03 [+0.01, +0.11]	+0.03 [+0.02, +0.11]	-0.03 [-0.07, +0.10]	

preprocessing examines every supplied crop, drops cells smaller than 40 pixels, excludes centroids within 48 pixels of a crop boundary, requires an assigned nucleus of at least 20 pixels, and removes duplicate cells in overlapping crop interiors. After these rules and feature availability, 3,331 crops contain retained cells; 3,323 have both near and far pairs. The crop geometry and these selection rules define the estimand. CellPlex uses a different whole-slide processing pipeline and fixed split; its cell count and scores are not directly comparable to this subset.

Features and folds. PCA (128 components) is fitted on 60,000 cells sampled from training patients of each rotation; cell features are standardized with training statistics. Regions are 512×512 crops at 0.325 μm/pixel; all cells retained by quality control are used, and pairs are computed among cells of the same region.

LIGATURE**.** A preliminary generator trained with one energy score on the full statistic vector left the attenuation unchanged (−34%), which motivated per-statistic scoring. Transformer width 128, 3 layers, 4 heads, attention bias from 16 Gaussian radial basis functions of distance (centres 0–160 μm, width 10 μm). Loadings are per-cell outputs of a two-layer head ($p \times K$, $K = 4 \cdot 4 + 8 + 4 + 4 = 32$), scaled by 0.3 at initialization. Fields and region latents are independent between regions; only the patient latent couples regions. Random Fourier features: 64 per field and region, frequencies $\mathcal{N}(0, \ell^{-2} I)$, redrawn for each realization. Conditional on these features, residual draws are Gaussian; integrating their randomness gives a finite-feature Gaussian mixture with the stated kernel covariance. The cell term uses a moment-matched Gaussian approximation; region and patient CRPS score the actual generated statistics. Gaussian marginal/likelihood statements refer to the ideal field or their specified Gaussian surrogate, not the integrated finite-feature likelihood. Batches: 4 patients × 4 regions, 16 joint samples. Variance calibration: Adam (learning rate 0.05, 4 epochs) on the seven log-scales with the network frozen. The 4-region patient average in training is a noisier target than the all-region average evaluated at test time; with 8 regions per patient at the same batch size, results were unchanged (MSE 0.90 vs 0.92 of the plug-in's, raw coverage 73% vs 71%). Field rank 8 left validation loss unchanged and was not adopted.

Analytic point forecasts. These remove Monte Carlo error from the conditional mean without changing the model; substituting expected moments into a ratio is a separate approximation. For bilinear statistics, conditional means follow by replacing each product by $f_{ia} f_{jb} + \mathrm{Cov}(Y_{ia}, Y_{jb} \mid H)$ and averaging with its bin weights. This integrates all latent and random-feature second moments without Monte Carlo error. The core release additionally returns these 832 expected coordinates. Reference and reseed results above retain their original 64-draw means; samples remain necessary for their predictive laws and nonlinear queries.

Direct contact model. The direct graph model's backbone (frozen embeddings, the stain's per-cell predictions as inputs, the same cell-graph transformer, mean and max pooling with log cell count) predicts, per region and per cell, the 16 positive-cell fractions π_a and the 256 conditional contact probabilities c_{ab}, each a logit-scale correction of the quantile-matched plug-in's (training-patient gates for the measured targets, validation-fitted gates for the plug-in), so predicted contacts $\pi_a c_{ab}$ never exceed predicted positives π_a. It

is trained with Gaussian CRPS on these per-cell counts for regions and patient averages (15 epochs, early stopping on validation patients, three seeds); a patient's contact fraction is the ratio of its summed predicted counts. The target-matched variant adds an absolute-error loss on each patient's pooled contact fraction (weight 10) and early-stops on it; it is the stronger baseline. An unconstrained variant predicting numerators and denominators separately is worse (ORION 0.82, lung 0.95).

Lung cohort. HEST-1k v1.3 Xenium samples NCBI856–NCBI884 of Vannan et al. [25] (20 samples, 19 patients; 0.2125 μm/pixel H&E aligned to the transcripts). Transcripts with quality ≥ 20 assigned to a Xenium cell are counted per gene; each cell is placed at the centroid of the nearest Xenium cell polygon (within 8 μm), whose area, perimeter and eccentricity are the morphology inputs, so no transcript-derived quantity enters the inputs. Targets are counts scaled to the median library size (cells with ≥ 10 transcripts). Genes: EPCAM, SFTPC, AGER, SCGB1A1, KRT5 (epithelial); PTPRC, CD3E, CD8A, MS4A1, CD68, MARCO, SPP1 (immune); COL1A1, ACTA2 (stromal); PECAM1 (endothelial); MKI67. Regions are 166.4 μm squares with at least 40 cells, 80 per sample; Phikon-v2 embeddings use the same 20.8 μm field of view (98 pixels). All model and training settings are those of ORION; contact statistics use the same 85th-percentile gates.

Virtual stains. *Transformer*: same architecture without noise, per-cell squared loss, 25 epochs, early stopping on validation patients. *Ridge*: closed form on standardized cell features, penalty from $\{10, \ldots, 10^6\}$ by validation MSE. *U-Net*: five-level encoder–decoder (32–384 channels) on H&E downsampled to 0.65 μm/pixel, pixel MSE plus per-cell MSE after mask pooling with a learned per-channel affine map, 40 epochs, early stopping on validation cell MSE. *ROSIE*: released ConvNeXt-Small weights, loaded with a restricted unpickler; one 148-pixel patch per cell centroid (48 μm, assuming 0.377 μm/pixel for ROSIE's training images), resized to 224, ImageNet normalization; one affine map per marker from the matched channel, fitted on training patients. *GigaTIME*: the released nested U-Net, re-implemented from the authors' description and loaded tensors-only; regions resized to 336×336 ($\approx$0.5 μm/pixel), per-pixel probabilities averaged over each cell mask; a ridge head on the 21 channel logits per marker, penalty chosen on validation patients. *GigaTIME fine-tuned*: the same network with its output layer replaced by 16 channels (rows of matched markers initialized from GigaTIME's), trained end to end with the U-Net's loss, AdamW (learning rate 3×10^{-4}, one-cycle schedule), 30 epochs, best validation cell MSE. Table 7 reports per-cell transfer.

Table 7: Per-cell Pearson correlation between published-model predictions and measured ORION intensities on test patients (mean over rotations). GigaTIME is shown for markers with a matched channel.

Marker	ROSIE	GigaTIME
CD163	0.15	–
CD20	0.25	0.22
CD31	0.19	–
CD3e	0.28	0.28
CD4	0.18	0.32
CD45	0.36	–
CD45RO	0.37	–
CD68	0.17	0.24
CD8a	0.21	0.21
E-cadherin	0.14	–
FOXP3	0.12	–
Ki67	0.08	0.43
PD-1	0.22	0.30
PD-L1	0.24	0.24
Pan-CK	0.25	0.63
SMA	0.14	–

Table 8: LIGATURE wrapped around six virtual stains (40 held-out patients). Cell MSE: per-cell test MSE of the stain. Bias: median relative bias of cohort means. MSE: patient-level MSE relative to the stain's own plug-in for all statistics and the 16 clustering statistics. CRPS: ratio to a calibrated Gaussian plug-in. Contact: patient-level MSE of gated contact fractions relative to the plug-in, for the quantile-matched plug-in and LIGATURE. †GigaTIME: bias, MSE and CRPS columns use the 45 statistics among the 9 markers it measures directly; contact uses all pairs, with other markers predicted by its linear head.

	cell	bias		MSE / plug-in		CRPS	contact MSE	
Virtual stain	MSE	plug-in	LIGATURE	all	same	LIGATURE/Gauss.	quant.	LIGATURE
Cell-graph transformer	0.78	−34%	−6%	0.88	0.60	0.84	0.88	0.62
Per-cell ridge, no context	0.85	−39%	−2%	0.86	0.64	0.84	0.83	0.72
Pixel U-Net	0.78	−12%	+3%	1.04	0.84	0.90	0.94	0.62
ROSIE	1.04	−94%	+5%	0.61	0.45	0.66	1.71	0.47
GigaTIME†	0.95	−68%	−6%	0.73	0.52	0.72	0.63	0.40
GigaTIME fine-tuned	0.78	−7%	+8%	0.96	0.81	0.89	0.94	0.67

A3 Additional results

A3.1 Additional gates and unscored query families

The advantage holds at 75th- and 95th-percentile gates (contact error 0.83 and 0.43 vs 1.09 and 0.91 for the quantile-matched plug-in). Three further families behave similarly (Appendix Fig. 5a): near-minus-far co-positivity (MSE 0.70 of the plug-in's), products at radii finer than the training bins (attenuation 39% to 10%; MSE 0.91), and positive-cell fractions (MSE 0.88, but bias +11% vs +5% for the quantile-matched plug-in, a tail limit of the Gaussian residual).

A3.2 Empirical rationale and identification analysis

The identification calculations use Gaussian marginals matched to the model's per-cell means and variances and bound conditional expectations over their admissible couplings. For non-Gaussian marginals the same semidefinite programs give moment-based outer bounds; endpoint attainability with fixed non-Gaussian marginals is not asserted. Comparisons with realized assay statistics provide empirical context; they do not validate predictive coverage or identify an individual tissue's molecular configuration.

Plug-in attenuation. Across the 136 statistics, the plug-in underestimates the cohort-mean statistic by a median of 34% (30% for cross-marker and 58% for same-marker statistics; Fig. 3a, Table 1). The direction matches Proposition 1: positive residual covariance between neighboring cells, which the per-cell mean cannot express, is missing (Fig. 1). The errors are not independent across regions: the plug-in error of a region statistic has a median within-patient intraclass correlation of 0.087 (interquartile range 0.057–0.107), a median design effect of 6.4 at 63 regions per patient (Proposition 5). Per-statistic calibration removes the cohort-mean bias on validation patients but transfers poorly: its patient-level MSE is 13% *higher* than the plug-in's and its rank correlation with the measured values is lower (0.41 vs 0.52), because affine corrections fitted on 8 patients do not transfer across patients.

What the stain does not capture. The fraction of a cohort-mean statistic missing from the stain ranges on ORION from 11–15% (epithelial–stromal segregation) to 66–79% (immune and vascular clustering; Appendix Fig. 6), and LIGATURE estimates it from H&E on held-out patients ($r = 0.90$ across 99 statistics).

Identification from model-implied marginals. We computed Proposition 2's interval for every same-marker statistic in 400 held-out regions (10 per test patient), using Gaussian marginals matched to LIGATURE's calibrated per-cell means and variances. The solver is certified (dual gap at most 0.33% at upper and 4.9% at lower endpoints; Appendix). Patient-level intervals (the average of region intervals, since couplings across regions are unconstrained) bound the conditional expectation; they contain the measured value for 97% of patient–marker pairs (all within two standard errors of the 10-region average) but are uninformative: their median width is 2.4 times the 5–95% range of the statistic across the 40 patients and exceeds it for 14 of 16 markers (all but Pan-CK and E-cadherin clustering). The extremes are attained by sign-pattern couplings that real tissue need not exhibit, so we also restricted the coupling to stationary, nonnegative correlations

decaying with distance. The restricted interval is narrower (median 0.64 of the cohort range) and, allowing for sampling error, consistent with the data (3.6% of measured values lie more than two standard errors outside it), yet it still spans most of the between-patient range. Per-cell predictive laws thus leave spatial statistics largely undetermined unless the dependence is assumed or, as in LIGATURE, learned.

Table 9: Per-marker identification intervals for clustering statistics (patient level). Cohort range: 5–95% range of the measured statistic across patients. Per-cell interval: width of Proposition 2's interval from calibrated per-cell laws; restricted: stationary, nonnegative, decaying correlations. Ratios are to the cohort range.

Marker	cohort	LIGATURE	per-cell	ratio	restricted	ratio
E-cadherin	0.55	0.30	0.34	0.6	0.09	0.17
Pan-CK	0.46	0.29	0.38	0.8	0.10	0.22
CD20	0.65	0.21	0.84	1.3	0.22	0.34
Ki67	0.46	0.31	0.86	1.9	0.23	0.51
CD4	0.61	0.26	1.19	1.9	0.32	0.52
CD45	0.40	0.27	0.82	2.0	0.21	0.53
CD8a	0.47	0.22	0.95	2.0	0.27	0.58
CD45RO	0.39	0.20	0.91	2.3	0.24	0.62
CD31	0.34	0.28	0.84	2.4	0.23	0.66
CD3e	0.43	0.25	1.30	3.0	0.35	0.81
PD-L1	0.21	0.12	0.66	3.2	0.17	0.82
SMA	0.35	0.27	1.24	3.6	0.31	0.89
CD163	0.34	0.34	1.24	3.7	0.32	0.95
PD-1	0.23	0.20	0.95	4.1	0.25	1.07
FOXP3	0.16	0.16	0.68	4.3	0.19	1.19
CD68	0.28	0.21	1.22	4.4	0.32	1.14

Table 10: Coverage of patient-level intervals by rotation (all 136 statistics).

Rotation	raw 90%	inflated	conformal	$\hat{\tau}$ infl.	$\hat{\tau}$ conf.
0	0.87	0.94	0.98	1.33	2.84
1	0.67	0.87	0.95	1.57	3.30
2	0.74	0.92	0.99	1.73	3.88
3	0.70	0.96	0.99	2.32	5.51
4	0.74	0.86	0.95	1.53	3.74

Table 11: Hybrid inference with measured anchor markers: patient-level MSE relative to the anchor-substituted plug-in, statistics involving at least one unmeasured marker.

Anchors	ridge imput.	LIGATURE, substituted	exact cond.	regularized cond.	δ
Pan-CK, CD45	1.07	0.90	1.36	0.95	1
PD-L1, CD8a	1.30	0.87	3.96	1.28	1
CD3e	1.17	0.88	1.47	0.94	1

Empirical bias identity. With $r = Y - f$ and mean error $\delta = \mu - f$, every binned moment splits as $G(Y) = G(f) + G(r) + X(f, r)$, where $\mathbb{E}[G(r) \mid H]$ is the residual second moment, $G(\delta)$ plus Proposition 1's covariance term, and $\mathbb{E}[X(f, r) \mid H] = X(f, \delta)$ vanishes when $f = \mu$. On test patients (Table 14) the residual second moment is similar for all in-domain stains (31–42% of the measured statistic), but the cross term is near zero only for the transformer (+3%; +4% on the lung cohort), the per-cell ridge and ROSIE (both −2%). For the pixel-level stains it is negative (−19%, −25%): their mean error is anti-aligned with neighboring predictions and offsets the residual second moment, which a frozen-mean wrapper cannot undo.

Table 12: Gated contact fractions for selected pairs across virtual stains: patient-level MSE relative to each stain's plug-in, quantile-matched plug-in / LIGATURE. Over all pairs, wrapping the ridge stain gives the same ordering (MSE 1.00, 0.83, 0.72; ρ 0.47, 0.50, 0.56).

Pair	transformer	ridge	U-Net	ROSIE	GigaTIME
CD8a → Pan-CK	0.99 / 0.60	0.57 / 0.52	0.88 / 0.68	2.60 / 0.43	0.67 / 0.32
CD3e → Pan-CK	0.82 / 0.43	0.37 / 0.41	0.67 / 0.51	6.11 / 0.71	0.85 / 0.40
CD68 → PD-L1	1.03 / 0.51	1.27 / 0.93	1.64 / 0.55	1.02 / 0.39	0.83 / 0.41
PD-L1 → CD8a	1.09 / 0.96	0.72 / 0.93	0.71 / 0.64	0.28 / 0.46	0.55 / 0.66
FOXP3 → CD8a	1.09 / 0.80	0.46 / 0.69	0.55 / 0.68	0.74 / 0.83	0.45 / 0.70
CD8a → CD68	0.69 / 0.33	0.32 / 0.18	0.81 / 0.39	0.14 / 0.08	0.23 / 0.09

Table 13: Cellular-neighborhood composition across virtual stains (40 held-out patients): mean total-variation distance between predicted and measured per-patient frequencies of 10 neighborhoods, Wilcoxon p and number of patients for which LIGATURE-Q is closer than the quantile-matched plug-in, and mean per-neighborhood rank correlation across patients (k-means is fitted per rotation; clusters are aligned to the first rotation's by Hungarian matching of centroids before pooling). GigaTIME markers without a matched channel come from its linear head.

	total variation to measured						rank corr.	
Virtual stain	plug-in	quant.	LIGATURE	LIGATURE-Q	p	better	quant.	LIGATURE-Q
Cell-graph transformer	0.42	0.34	0.36	0.29	0.005	26/40	0.52	0.57
Per-cell ridge	0.54	0.33	0.42	0.29	0.006	27/40	0.54	0.54
Pixel U-Net	0.44	0.34	0.36	0.28	0.002	30/40	0.54	0.55
ROSIE	0.72	0.54	0.46	0.36	4.6×10^{-10}	36/40	0.18	0.36
GigaTIME	0.69	0.40	0.44	0.32	3.8×10^{-5}	32/40	0.37	0.43
GigaTIME fine-tuned	0.49	0.36	0.42	0.29	5.7×10^{-4}	29/40	0.51	0.51

Table 14: Empirical bias identity on test patients. For residual $r = y - f$, each binned moment satisfies $G(y) = G(f) + G(r) + X(f, r)$; $G(r)$ is the residual second moment (Proposition 1's covariance term plus the stain's mean-error term $G(\delta)$, $\delta = \mu - f$) and X the predictor–residual cross term, zero in expectation when f is the conditional mean μ. Columns: median over the statistics with |measured cohort mean| > 0.02 (99 on ORION, 62 on the lung cohort) of the plug-in's relative bias and of each term's cohort mean relative to the measured cohort mean; share of statistics with negative cross term; median per-patient root-mean-square of each term relative to the between-patient standard deviation. GigaTIME is evaluated here on all 136 statistics (other markers from its linear head), so its bias differs from Table 8.

	plug-in	cohort mean / measured			per-patient RMS / s.d.	
Virtual stain	bias	$G(r)$	$X(f,r)$	$X<0$	$G(r)$	$X(f,r)$
Cell-graph transformer	−34%	+31%	+3%	39%	0.47	0.64
Per-cell ridge	−39%	+42%	−2%	61%	0.60	0.65
Pixel U-Net	−12%	+34%	−19%	93%	0.53	0.70
ROSIE	−94%	+90%	−2%	54%	1.09	0.34
GigaTIME	−59%	+71%	−14%	84%	0.89	0.60
GigaTIME fine-tuned	−7%	+36%	−25%	88%	0.56	0.75
Lung (Xenium), transformer	−71%	+64%	+4%	32%	0.74	0.57

Table 15: Patient-disjoint calibration on ORION (5 rotations, 40 test patients). Stain and generator are retrained on 20 patients per rotation; 2 patients serve model selection and variance fitting and 10 serve calibration only. Per-statistic split-conformal 90% intervals (rank 10 of 10); width and interval score are normalized per statistic and averaged; calibration does not change point predictions (clustering MSE 0.66 of the matched plug-in's). Coverage is marginal per statistic, not simultaneous or patient-conditional; ORION was examined during development, so this is not a confirmatory cohort. An earlier exact version with only 4 calibration patients (wrapping ROSIE, whose maps use training patients only) was uninformative even at an 80% target.

	clustering (16)			all 136		
Method	cov.	width	score	cov.	width	score
LIGATURE samples, raw	45.6%	1.06	6.37	70.4%	1.75	4.85
LIGATURE + conformal	90.0%	3.81	4.98	90.2%	3.59	4.86
LIGATURE center, constant scale + conformal	90.9%	3.77	5.22	90.9%	3.95	5.64
plug-in, Gaussian (validation scale)	74.7%	2.72	7.05	72.0%	2.39	6.78
plug-in + conformal	91.1%	4.63	6.69	91.5%	4.09	5.89
global rescale + conformal	92.3%	4.47	6.18	90.6%	4.88	7.13
family rescale + conformal	91.7%	4.40	6.09	90.8%	4.92	7.13
rescale, LIGATURE scale + conformal	92.3%	4.46	5.76	89.5%	4.17	5.67

Table 16: Patient-disjoint calibration of statistics never scored in training (same refit as Table 15; gates and neighborhood templates fitted on the 20 training patients, quantile-matched thresholds on the 2 selection patients, 10 calibration patients). Prespecified mean/SD conformal 90% intervals; width and interval score in the statistic's units. Width and score differences from the quantile-matched plug-in are not statistically significant (descriptive paired intervals cross zero), and a constant-scale interval around LIGATURE-Q scores similarly; point error is 26% (contacts) and 24% (neighborhoods) below the quantile-matched plug-in's, as ratios of equal-patient mean squared errors (each patient's squared errors averaged over its eligible statistics, then patients averaged), not per-statistic normalized ratios as elsewhere; coverage, width and score also give patients equal weight; calibration does not change point estimates. Contact pairs require at least 20 measured positive source cells, an outcome-defined eligibility, and 25% of adaptive contact intervals span all of $[0, 1]$. A conformalized-quantile variant chosen after these results (a development diagnostic, not prespecified) narrows neighborhood intervals by 12% relative to the quantile-matched plug-in (better in 38 of 40 patients).

	contact fractions			neighborhood frequencies		
Method	cov.	width	score	cov.	width	score
LIGATURE-Q samples, raw	45.5%	0.247	1.656	44.3%	0.089	0.748
LIGATURE-Q + conformal	91.6%	0.646	0.808	92.8%	0.321	0.421
LIGATURE-Q center, constant scale + conformal	92.5%	0.642	0.798	92.2%	0.290	0.421
LIGATURE + conformal	91.4%	0.613	0.783	90.0%	0.316	0.447
quantile-matched plug-in + conformal	92.6%	0.653	0.819	90.0%	0.310	0.461
plug-in + conformal	91.9%	0.633	0.846	88.5%	0.311	0.478

Table 17: Patient-disjoint calibration on the lung cohort (5 rotations, 19 test patients): stain and generator retrained on 10–11 patients, 1 for model selection and variance fitting, 4 for calibration only. With 4 calibration patients a finite split-conformal interval exists only up to 80%, so the target (fixed before fitting) is 80% (rank 4 of 4), not 90%. Per-fold clustering coverage ranges 65.6–90.6%; the constant-scale control is narrower with a similar score, so adaptive spread shows no established advantage here; one selection patient makes rescaling baselines unstable. Development evidence; point MSE 0.49 (clustering) of the refit's matched plug-in.

	clustering (16)			all 136		
Method	cov.	width	score	cov.	width	score
LIGATURE samples, raw	27.6%	0.57	4.78	50.2%	1.29	4.50
LIGATURE + conformal	82.6%	3.57	4.60	80.3%	3.38	4.84
LIGATURE center, constant scale + conformal	81.6%	3.05	4.60	80.1%	3.36	5.18
plug-in, Gaussian (validation scale)	55.9%	3.07	6.84	49.4%	1.71	5.95
plug-in + conformal	79.9%	4.44	6.77	79.6%	3.58	5.69
global rescale + conformal	79.3%	4.01	6.07	77.1%	4.39	7.59
family rescale + conformal	78.3%	4.13	6.52	78.2%	3.81	6.28
rescale, LIGATURE scale + conformal	80.6%	4.50	5.80	77.6%	4.42	6.58

Mask-exclusion protocol. Before computing sensitivity metrics, we fixed the rule in the released analysis plan: exclude near pairs if pixels in their supplied whole-cell masks have Chebyshev distance at most two pixels (0.65 μm along an axis); keep the far bin unchanged. This removes 392,792 of 1,938,513 undirected near pairs (20.3%) and leaves 1,545,721. Distance bins use the original stored masks, avoiding threshold changes from float32 distance rounding. One originally eligible region has no retained near pairs and is excluded from both diagnostic variants. The stain and generator are unchanged, with each patient's original held-out model; no control-specific training or threshold tuning occurs. Conditional expected products are $f_{ia}f_{ja} + \hat{\Sigma}_{ia,ja}$, integrated over the sampled Gaussian fields and their random Fourier features. A 1,024-draw check on the first held-out region agrees within Monte Carlo uncertainty. The restricted target differs from the original statistic. This is a post-development sensitivity check, not intensity-spillover correction, a segmentation benchmark or a claim that remaining dependence is biological.

Table 18: All estimators on one scale: patient-level MSE relative to the cell-graph transformer's plug-in, averaged over the 136 statistics (all) and the 16 clustering statistics (same), and mean Spearman correlation with measured values across 40 held-out patients. No stain: estimators that use only the cell embeddings. Learned mean correction: LIGATURE with a zero-initialized correction added to the frozen stain. Direct graph with stain inputs: the direct model given the stain's per-cell predictions, predicting a correction of the stain's plug-in statistics. Conditional Gaussian NLL: the backbone trained by the joint Gaussian likelihood of each region's cells (Woodbury form, one random-feature draw per region), early-stopped on validation likelihood, conditional on the sampled features, with a $0.01I$ cell-covariance nugget for invertibility that is absent from the sampler; this is not the integrated mixture likelihood. Results precede variance calibration (calibrating its scales by likelihood made it worse; on the lung cohort the fit was numerically unstable).

Stain	Estimator	MSE all	MSE same	ρ
No stain	Direct ridge on region features	1.42	0.99	0.31
	Direct graph (no stain)	1.30	0.91	0.38
	LIGATURE around a constant stain	1.13	0.76	0.46
	LIGATURE around a constant stain, learned mean	1.00	0.75	0.48
Transformer	plug-in	1.00	1.00	0.52
	global rescale	0.89	0.83	0.54
	Direct graph (stain inputs)	0.89	0.63	0.53
	regularized conditional Gaussian NLL	2.75	4.46	0.35
	LIGATURE	0.88	0.60	0.53
Ridge	plug-in	1.36	1.32	0.42
	LIGATURE	1.13	0.78	0.42
U-Net	plug-in	1.07	0.98	0.51
	LIGATURE	1.09	0.76	0.47
	LIGATURE, learned mean correction	1.03	0.72	0.48
ROSIE	plug-in	2.37	2.37	0.18
	LIGATURE	1.17	0.87	0.46
GigaTIME	plug-in	1.80	1.86	0.28
	LIGATURE	1.14	0.79	0.43
	LIGATURE, learned mean correction	1.09	0.74	0.44
GigaTIME fine-tuned	plug-in	1.12	0.95	0.51
	LIGATURE	1.05	0.74	0.51
	LIGATURE, learned mean correction	1.06	0.72	0.51

Table 19: Known-law simulations on ORION's cells, regions, patients and embeddings (40 patients, five held-out rotations). Targets are $Y = f^\star + R$ with $f^\star$ the fold-0 stain's predictions, so the plug-in of $f^\star$ is the Bayes per-cell predictor, and R a member of LIGATURE's class: exact squared-exponential fields at 20 and 80 μm (rank 2) with embedding-dependent loadings, cell noise ($\sigma = 0.6$) and a patient latent. The second case adds a shared cell-level factor (loading 0.6) to CD8a/Pan-CK and CD68/PD-L1, a within-cell cross-marker covariance with zero bin averages. LIGATURE is fitted to the simulated targets with the reference configuration and $f^\star$ as its frozen stain. *Top*: patient-level MSE of the 136 contrasts relative to the plug-in and median relative bias; the Bayes statistic predictor adds the exact $\mathbb{E}[G(r) \mid H]$ under the simulated law to the plug-in. *Bottom*: residual correlations on test cells, within a cell and between cells 0–25 μm apart (normalized by within-cell variances), true and implied by LIGATURE's loadings.

	Correct model class			Within-cell dependence added		
Contrast estimator	MSE all	MSE same	bias	MSE all	MSE same	bias
Plug-in of the true mean	1.00	1.00	−38%	1.00	1.00	−37%
LIGATURE	0.59	0.12	+1%	0.61	0.12	+2%
Bayes statistic predictor	0.32	0.07	−1%	0.33	0.07	0%

	Correct model class				Within-cell dependence added			
	within cell		0–25 µm		within cell		0–25 µm	
Residual correlation	true	LIGATURE	true	LIGATURE	true	LIGATURE	true	LIGATURE
CD8a / Pan-CK	−0.14	−0.00	−0.12	−0.09	+0.22	+0.01	−0.09	−0.06
CD68 / PD-L1	+0.01	−0.11	−0.03	−0.01	+0.36	−0.20	−0.02	−0.01
Other 118 pairs, mean abs. error	0.12		0.02		0.12		0.01	

Table 20: Fixed-model same-marker sensitivity to immediately neighboring supplied cell masks (40 patients, 16 markers). Both variants use the same 3,322 regions, equal-region patient averages and each variant's own plug-in denominator. LIGATURE predictions integrate the model's second moments analytically, including random Fourier features; they are not the stored 64-draw mean used in Table 1. Intervals are 95% patient bootstraps conditional on the fixed models (2,000 replicates). Bias is the median signed relative cohort-mean error across the 16 markers with measured mean magnitude above 0.02.

Near-pair rule	MSE / plug-in [95% interval]	Plug-in bias	LIGATURE bias
All near pairs	0.60 [0.54, 0.72]	−58%	−5%
Non-touching masks	0.72 [0.62, 1.07]	−53%	+6%

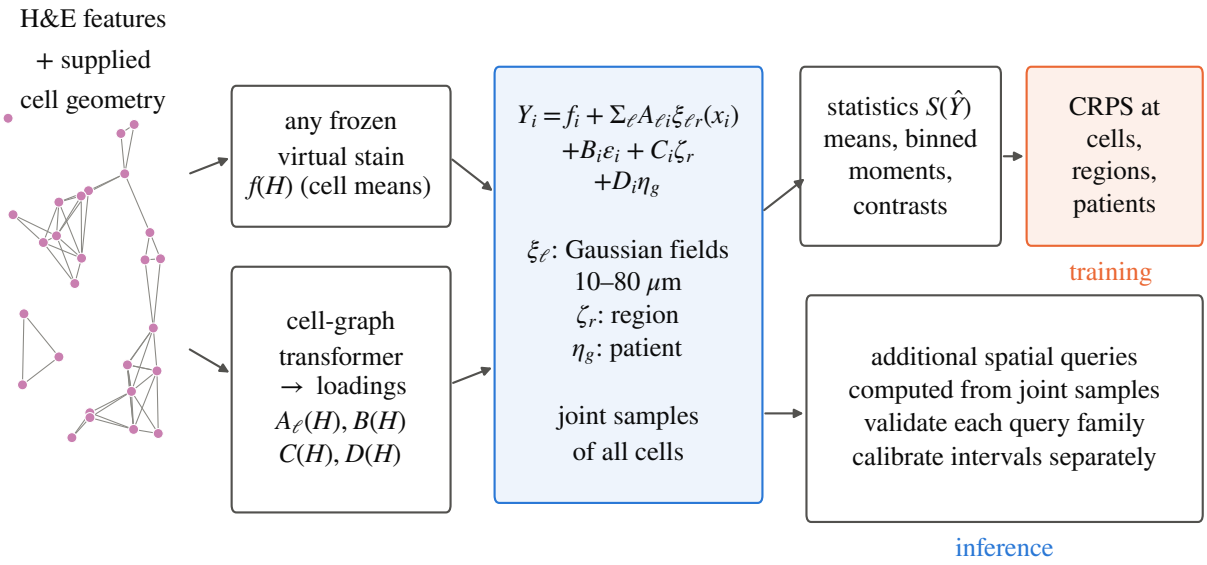


Figure 7: LIGATURE maps a frozen stain and cell features to a joint residual distribution, trained by CRPS on cell markers, region statistics and patient averages. Shared patient latents couple samples across regions. Primary geometry is supplied by the assay segmentation; additional computed queries require separate validation, and their intervals require query-specific calibration. An illustrative held-out region is shown in Appendix Fig. 4.

A4 Additional rationale and diagnostic commentary

Proposition 7 (Likelihood versus statistic scoring)**.** *Let a residual model have covariance* $\Sigma_\theta = \sum_k \theta_k K_k$ *with fixed positive semidefinite components* K_k *(cell noise, fields at fixed scales, latents) and let the truth be* Σ^**. (i) At an interior optimum, the Gaussian-likelihood projection satisfies* $\operatorname{tr}\big(\Sigma_\theta^{-1} K_k \Sigma_\theta^{-1}(\Sigma^* - \Sigma_\theta)\big) = 0$ *for every free* θ_k*, whereas, for a frozen mean equal to the true conditional mean, a fit reproducing the means of the analyzed bilinear statistics* Q_{W_j} *solves* $\operatorname{tr}\big(W_j(\Sigma^* - \Sigma_\theta)\big) = 0$ *for every* j*: a* Σ^* *in the class satisfies both sets of equations, without implying uniqueness for the statistic equations; under misspecification likelihood weighs covariance errors by the whitened component designs and statistic scores by the analysis weights. (ii) For a positive-definite unit-variance covariance with correlation* $\rho \geq 0$ *on the* N_1 *cell pairs of bin* B_1*, fitted with* $\{I, \mathbf{1}\mathbf{1}^\top\}$*, likelihood gives region-latent variance* $c^2 = \rho N_1 / \binom{n}{2}$*, attenuating the bin-1 covariance by the fraction of pairs outside* B_1*, and the bin-1 statistic gives* $c^2 = \rho$*.*

In ORION the plug-in errors of the 136 region statistics have a median within-patient intraclass correlation of 0.087 (interquartile range 0.057–0.107); with a median of 63 regions per patient and this ICC as a proxy for $\bar{\rho}$, the median design effect is about 6, so region-independent uncertainty would understate patient-level variance several-fold. LIGATURE therefore shares a patient latent across regions and scores patient averages of the statistics, for which Proposition 3 applies unchanged.

A4.1 Downstream checks

Association studies. For the 36 pairs of nine predefined spatial biomarkers, we estimated across-patient Spearman correlations as a virtual-population study would (Appendix Fig. 5b). LIGATURE, recomputing each correlation per joint sample, has mean absolute error 0.099 versus 0.149 for the plug-in; the difference is not statistically significant with 40 patients (bootstrap 90% interval for the reduction −2% to 40%, including zero); its intervals cover the measured correlation for 83% of pairs versus 78% for Fisher intervals on the plug-in.

Which patients to measure. Measuring the 20% of patients with the widest predictive distribution removes 50% of the cohort's squared error (random 20%, oracle 80%; Appendix Fig. 5c), but, as reported for virtual transcriptomics [30], this signal is mostly carried by the predicted value: ranking by the magnitude of the plug-in statistic removes 51%, and the part of the spread not explained by LIGATURE's predicted magnitude has partial rank correlation 0.18 with the error (28% removed). Predictive spread adds little triage information beyond the predicted magnitude here.

A4.2 Ablation detail

Systematic ablations (36 variants of five rotations on both cohorts; Appendix Table 5) separate what matters. The three current-code reseeds have MSE 0.92 ± 0.02 (range 0.898–0.945; clustering 0.60–0.63), with raw coverage $72.9 \pm 0.9\%$ (range 72.1–73.9%); small ablation differences should be assessed against this observed variation. The rerotation, with the stain retrained, gives 0.93 (clustering 0.64; per-patient Wilcoxon $p = 0.07$ and 5×10^{-6} against the plug-in), median bias $-34\% \to -5\%$, while the direct graph (stain inputs) has 0.91 overall and 0.69 for clustering. Essential: statistic-level scores (scored per cell only, as in engression-style generators [16], MSE 1.33 with current code versus 2.40 in the earlier Table 4 run; lung 2.27), scoring the bin contrasts (1.29 without), morphology-dependent loadings (stationary: clustering 0.67, raw coverage 39%) and the patient latent (current raw coverage 49.2% without; earlier run 49.8%). Spatial fields reduce overall error (1.00 without them with current code, 1.03 in the earlier run; lung 0.94 vs 0.84–0.86), but without them ORION clustering (0.62) and CRPS (0.0282) stay within seed variation, with smaller bias magnitude (+2%) and higher raw coverage (82%). Within noise: any one statistic-level score, the region latent, the number of field scales or their rank, network depth, morphology descriptors, the stain as residual-network input, and 6 rather than 24 training patients; a narrower network or 5 epochs scored at the favorable edge, so the reference is not tuned to its limit. Independently sampled Gaussian per-cell residuals with fitted marginal variances leave the attenuation unchanged and cover 28% (Table 4), as Proposition 2 predicts.

Table 21: ORION evaluation labels used throughout. Patient roles are training/validation/test for 24/8/8, and training/selection/calibration/test for 20/2/10/8. Runs (L/D) count LIGATURE/direct fits per rotation for the scored-statistic head; new-query direct heads are fitted separately. MSE is overall patient-level MSE relative to that evaluation's plug-in. Reseed variation is sample SD. The four-run aggregate combines the reference run with three reseeds; its paired direct comparison uses the direct reseed mean. Independent calibration and H&E geometry are separate evaluations; H&E geometry also changes the cell-proxy target.

Evaluation	Patient roles	Runs (L/D)	LIGATURE MSE	Direct graph MSE
Reference run	24/8/8	1/1	0.88	0.89
Reseed mean	24/8/8	3/3	0.92±0.02	0.88
Four-run aggregate	24/8/8	4/–	0.91	–
Rerotation	24/8/8	1/1	0.93	0.91
Independent calibration	20/2/10/8	3/3	0.95	0.96
H&E geometry	20/2/10/8	1/1	0.96	0.93

A5 Matched forecast, query-reuse and geometry checks

Fixed protocol and scope. These follow-up experiments use the five independent-calibration rotations (20 training, 2 selection, 10 calibration and 8 test patients). Training/selection features and the frozen stains reproduce that refit. Seeds 271027, 272027 and 273027 are fixed for the residual generators and direct heads; the first generator seed is reused. Stains are fixed across these seeds, so this does not measure full-pipeline seed variability. Global/family rescaling fitted on the two selection patients is unstable here (overall MSE ratios 1.85/1.84), unlike the eight-validation-patient reference protocol. The direct models use the same transformer inputs, stain predictions, width/depth, region/patient CRPS, 15 epochs and selection rule. One head predicts all 832 scored coordinates; another jointly predicts the 48 new-query coordinates. No calibration or test labels enter model fitting, early stopping, variance rescaling or ridge selection. These are repeatedly examined development cohorts, not preregistered confirmation. Paired bootstrap intervals resample 40 patients conditional on all fits and do not include all cross-fold training dependence. Only the two primary new-query tests use Holm correction; other comparisons are exploratory.

Queries and eligibility. For the symmetric binary adjacency $A_{ij} = \mathbf{1}\{0 < d_{ij} \le 25\,\mu\text{m}\}$, with no row normalization, let $W = \sum_{ij} A_{ij}$ and $z_i = Y_i - \bar{Y}$. For each marker,

$$I = \frac{n}{W}\frac{\sum_{ij} A_{ij} z_i z_j}{\sum_i z_i^2}, \qquad C = \frac{n-1}{2W}\frac{\sum_{ij} A_{ij}(Y_i - Y_j)^2}{\sum_i z_i^2}.$$

We average these region statistics equally within each patient. All sixteen markers are evaluated; Moran and Geary are related autocorrelation summaries, not independent replications. The secondary log-assay contrast inverts training-only marker standardization before taking near-minus-far products. Fold-specific centering changes that estimand on an irregular graph: near- and far-weighted cell means need not coincide. Training-standardized contrast improvements therefore do not imply equal improvements on the original intensity scale.

Every role requires nonempty near and far graphs. A selection-only audit found one two-cell crop with no admissible pairs entering the initial new-query fit. Before inspecting test performance, we excluded this crop consistently, centered products before summation for numerical stability, and reran affected query heads/forecasts. Original scored-statistic targets and valid direct-statistic fits are unchanged; prior outputs and the repair audit are retained. Constant-marker queries use an explicit zero convention, but no such measured region-marker case occurs in the eligible cache.

Readouts and dependence controls. Ridge uses 569 frozen image/stain context features, including region feature means/standard deviations, frozen stain context, cell count and plug-in queries; one penalty from $10^{-4}, \ldots, 10^{4}$ is selected jointly on the two selection patients. The affine baseline fits each query on selection regions. Neither requires additional assays: new targets are computed from existing training measurements. Frozen direct-moment reuse combines predicted means and near-bin moments, selection-cell residual variance

Table 22: New-query point error relative to each query's plug-in, three seeds and all 40 patients. Smaller is better. †Post hoc (Appendix A5.1). Adapted methods jointly learn 48 outputs from existing paired training assays. Frozen methods use no new-query labels for point fitting (separate measured calibration remains necessary for intervals). Moment ratios approximate expectations of ratios; log-assay moments are bilinear and integrated exactly.

Method	Moran I	Geary C	Log-assay contrast
Plug-in	1.000	1.000	1.000
No-image training mean†	0.423	0.515	2.805
Plug-in + constant†	0.315	0.377	1.376
Selection affine	0.318	0.403	1.526
Frozen-feature ridge (adapted)	0.207	0.285	0.954
Direct graph (adapted)	0.272	0.351	0.984
Frozen direct-moment reuse	8.334	10.070	0.966
Independent Gaussian cells	2.245	3.159	1.000
Exact independent cell marginals	2.245	3.161	1.005
LIGATURE moment ratios	0.559	0.526	0.956
LIGATURE joint samples	0.427	0.459	0.981

and a constant cell–regional-mean covariance approximation; variance projections and graph-theoretic ratio bounds keep estimates admissible. It performs poorly, so failure of this approximation is not evidence against all possible direct-model reuse. The joint-moment control integrates the fitted generator's second moments before forming approximate ratios. Independent Gaussian cells use the same marginal means/variances. A stronger independent-cell control preserves the entire within-cell marker-vector law of the finite-feature generator, including its field-induced Gaussian scale mixture, while independently sampling every cell. Both lose autocorrelation accuracy (Table 22).

Point differences and interval controls. Against the adapted direct graph, sample-based relative-MSE differences for Moran and Geary are $+0.156$ (95% interval $[0.071, 0.245]$) and $+0.108$ ($[0.026, 0.188]$), favoring the direct model. Holm-adjusted two-family p-values are 0.0029 and 0.0094. For scored clustering, the difference is -0.083 ($[-0.209, 0.029]$); its apparent advantage is not statistically significant. The overall interval-score improvement in Table 2 is driven mainly by cross-marker forecasts (difference -0.603, $[-0.884, -0.322]$). Overall raw normalized CRPS is 0.467 for LIGATURE versus 0.487 for the direct model; the paired difference interval $[-0.046, 0.004]$ includes zero. A Gaussian spread multiplier fitted only on the two selection patients worsens direct CRPS to 0.497, so it is not used as the sole comparator. Full raw, adaptive and constant-scale interval results follow. Analytic means change the center only (MSE ratios 0.94 overall and 0.66 for clustering, with the coverage, width and score in the table below); their reported CRPS and raw quantiles retain the sampled law.

Interval construction. For each coordinate, ten calibration patients supply the tenth order statistic of absolute residual divided by predicted spread; this is the finite-sample rank for nominal 90% split-conformal intervals. Constant-scale intervals use unit spread with the identical center and patients. Coverage is marginal for each statistic, not simultaneous over a patient's queries. Width and interval score divide by the training-patient standard deviation of that coordinate, preventing marker scale from dominating the forecast summary. Raw quantiles remain substantially under-covered. Adaptive scaling helps the scored cross-marker family but hurts the new Moran/Geary families relative to their own constant-scale controls, another limit on transferring uncertainty to unseen queries. Rows marked † are post hoc (Appendix A5.1).

Family	Method / interval	Coverage	Width	Score
All 136	Direct / raw	72.3%	1.37	5.73
All 136	Direct / adaptive	90.9%	3.75	5.57
All 136	Direct / constant	91.0%	3.91	5.73
All 136	LIGATURE / raw	70.2%	1.72	4.91
All 136	LIGATURE / adaptive	90.5%	3.72	4.98

Family	Method / interval	Coverage	Width	Score
All 136	LIGATURE / constant	91.1%	3.99	5.67
All 136	LIGATURE analytic / raw	70.2%	1.72	4.91
All 136	LIGATURE analytic / adaptive	90.5%	3.71	4.97
All 136	LIGATURE analytic / constant	91.2%	3.99	5.65
All 136	No-image mean† / constant	89.8%	4.62	6.78
All 136	Plug-in + constant† / constant	90.9%	3.89	5.58
All 136	Stain-context ridge† / constant	90.9%	4.01	5.80
Clustering	Direct / raw	69.9%	1.22	6.00
Clustering	Direct / adaptive	90.7%	3.78	5.65
Clustering	Direct / constant	91.4%	3.94	5.88
Clustering	LIGATURE / raw	44.8%	1.06	6.40
Clustering	LIGATURE / adaptive	90.2%	4.06	5.16
Clustering	LIGATURE / constant	91.6%	3.91	5.29
Clustering	LIGATURE analytic / raw	44.8%	1.06	6.40
Clustering	LIGATURE analytic / adaptive	90.2%	4.05	5.14
Clustering	LIGATURE analytic / constant	91.8%	3.90	5.28
Clustering	No-image mean† / constant	90.2%	4.51	6.74
Clustering	Plug-in + constant† / constant	92.2%	3.78	5.39
Clustering	Stain-context ridge† / constant	91.1%	3.91	5.56
Cross-marker	Direct / raw	72.6%	1.39	5.69
Cross-marker	Direct / adaptive	91.0%	3.75	5.56
Cross-marker	Direct / constant	90.9%	3.90	5.71
Cross-marker	LIGATURE / raw	73.6%	1.81	4.71
Cross-marker	LIGATURE / adaptive	90.5%	3.67	4.96
Cross-marker	LIGATURE / constant	91.0%	4.01	5.73
Cross-marker	LIGATURE analytic / raw	73.6%	1.81	4.71
Cross-marker	LIGATURE analytic / adaptive	90.5%	3.66	4.94
Cross-marker	LIGATURE analytic / constant	91.1%	4.00	5.69
Cross-marker	No-image mean† / constant	89.8%	4.64	6.79
Cross-marker	Plug-in + constant† / constant	90.7%	3.90	5.61
Cross-marker	Stain-context ridge† / constant	90.9%	4.02	5.83
Moran I	Direct / raw	68.0%	2.28	6.92
Moran I	Direct / adaptive	91.4%	4.45	5.63
Moran I	Direct / constant	90.6%	4.43	5.62
Moran I	LIGATURE / raw	34.4%	1.53	18.42
Moran I	LIGATURE / adaptive	90.3%	6.70	7.89
Moran I	LIGATURE / constant	91.6%	5.47	6.26
Moran I	No-image mean† / constant	89.5%	4.06	5.08
Moran I	Plug-in + constant† / constant	89.5%	4.42	5.88
Geary C	Direct / raw	72.7%	2.32	5.64
Geary C	Direct / adaptive	91.4%	3.93	4.73
Geary C	Direct / constant	90.8%	3.92	4.77
Geary C	LIGATURE / raw	37.2%	1.19	12.96
Geary C	LIGATURE / adaptive	90.4%	5.42	6.32
Geary C	LIGATURE / constant	91.5%	4.56	5.26
Geary C	No-image mean† / constant	90.2%	4.24	5.35
Geary C	Plug-in + constant† / constant	90.8%	3.89	4.86
Log-assay	Direct / raw	84.1%	1.82	3.45
Log-assay	Direct / adaptive	91.6%	2.75	3.57
Log-assay	Direct / constant	90.9%	2.85	3.70
Log-assay	LIGATURE / raw	68.6%	1.35	3.72
Log-assay	LIGATURE / adaptive	90.8%	3.15	3.81

Family	Method / interval	Coverage	Width	Score
Log-assay	LIGATURE / constant	91.7%	2.88	3.71
Log-assay	No-image mean† / constant	89.5%	4.20	5.75
Log-assay	Plug-in + constant† / constant	90.8%	3.02	3.94

Measured computation costs. On one eight-thread AMD EPYC 9654 allocation, 24 fixed test regions from eight patients take 0.355 s to generate 64 joint cell draws, plus 0.044 s to compute all 48 queries from cached draws. The adapted direct head takes 0.146 s including its plug-in queries. Ridge takes 0.052 s with query-independent context cached during the common stain forward pass (0.170 s recomputing that context). Frozen Phikon features and stain predictions are precomputed for all methods. These are illustrative single-allocation measurements, not a global throughput benchmark. Newly trained generator fits average 418 s, direct scored heads 237 s and direct query heads 229 s across the heterogeneous allocations; reused fits are not counted as zero-cost training. Cheap ridge/affine fitting takes 0.2–2.9 s per fold after feature preparation. Thus sample reuse avoids a new fit, but a general amortized accuracy/cost advantage over cheap adaptation is not established.

A5.1 Post hoc simple baselines

We added these baselines after test results had been seen; they are outside the fixed protocol above and are exploratory. None uses test labels or fits a neural model, and all use constant-scale intervals with the same ten calibration patients and conformal rank. The *no-image mean* is the training-patient mean of each statistic. The *plug-in plus constant* adds one shift per statistic. In the reference and lung evaluations, the shift is the mean of measured minus plug-in values over validation patients. In the 20/2/10/8 protocol two selection patients are too few (overall MSE ratio 1.50), so the shift is the training-patient measured mean minus the mean plug-in over selection and calibration patients, whose labels it does not use. Using the training patients' own in-sample plug-ins instead gives 0.93 overall and 0.68 for clustering, at score 5.57: lower overall point error than LIGATURE's 0.95, higher clustering error than its 0.66, and a worse interval score (LIGATURE difference -0.59, $[-0.92, -0.22]$).

The *stain-context ridge* is the frozen-feature ridge of Table 22, with the same 569 features, plug-in-residual target, penalty grid and joint selection on two patients, fitted to the 136 statistics. Selected penalties across rotations are 10^4, 0.01, 100, 100 and 100. Fixed penalties are a sensitivity check evaluated on test patients, not a selection rule: those from 10 to 10^4 reach MSE 0.93–0.94 (clustering 0.63–0.68), and smaller ones are worse (up to 1.56 overall). At the penalty with the lowest overall score (5.61), LIGATURE's difference remains -0.62 ($[-0.92, -0.32]$). The lowest clustering score across penalties is 5.13, against LIGATURE's 5.16 and a selected-penalty clustering difference of -0.40 ($[-0.98, 0.24]$).

Point differences from LIGATURE are inconclusive for the constant (-0.05, $[-0.18, 0.02]$; clustering -0.07, $[-0.20, 0.06]$) and the selected-penalty ridge (-0.07, $[-0.19, 0.01]$; clustering -0.04, $[-0.12, 0.11]$). Median bias in this protocol, overall and for clustering, is -16% and -26% for the direct graph, -6% and -2% for LIGATURE, $+5\%$ and $+3\%$ for the constant, and -9% and -9% for the ridge. On the lung cohort, LIGATURE's MSE differences from the constant are -0.13 overall ($[-1.05, -0.09]$) and -0.15 for clustering ($[-1.05, -0.06]$). Intervals are 95% paired patient bootstraps conditional on the fits (2,000 replicates).

A5.2 H&E-derived geometry and refitted models

We fix the first twenty non-overlapping crops per patient by ascending source coordinates before examining this control's targets or performance (789 crops, all forty patients). Selection uses the original assay-selected release, not an independent whole-slide sample. The official InstanSeg brightfield_nuclei v0.1.1 model [27] supplies nuclei from H&E alone. We resize 0.325 μm/pixel images to its 0.5 μm/pixel scale, use channel percentile normalization and default thresholds, then resize labels back by nearest-neighbor interpolation. Nearest-label expansion by six original pixels (1.95 μm) defines nucleus-plus-rim proxies. We exclude nuclei below 20 pixels, proxies below 40 pixels, zero-perimeter nuclei and centroids within 48 pixels of crop edges. This yields 79,600 proxies; one one-cell crop cannot form pair statistics. Source masks and assay-derived cell features are never read for model inputs.

Pinned Phikon-v2 features are extracted afresh from 64-pixel H&E fields centered at the new proxy centroids. Paired immunofluorescence supplies mean proxy intensities as targets only. The pretrained seg-

mentation model card, selected crop list and feature/model hashes accompany the analysis. The model SHA256 is `0c724da169d507d8dd2e206ad3675a0b1566824eee6e20ad9a0ba916311d9dd0`; weights come from `https://github.com/instanseg/instanseg/releases/tag/instanseg_models_v0.1.1`.

Stain, generator and matched direct graph are separately refitted in every 20/2/10/8 rotation with the original settings (25 stain epochs, 15 generator/direct epochs; seed 271027), no control-driven tuning, and independent calibration. Table 24 uses all forty held-out patients. The generator's overall point-error difference from the direct model is inconclusive (+0.035 in plug-in-normalized units, 95% interval $[-0.032, 0.099]$), as is the clustering difference (-0.020, $[-0.140, 0.079]$). Its clustering MSE relative to its own plug-in is 0.701 ($[0.623, 0.891]$); overall 0.960 ($[0.896, 1.116]$). These are single-seed, conditional patient bootstraps, not estimates of training or model-selection variability.

Relative to the direct adaptive model, the overall interval-score difference is -0.849 (95% interval $[-1.485, -0.269]$). For clustering it is -0.646, with interval $[-1.222, -0.192]$. Overall raw ensemble coverage is 58.0%, versus 90.6% after independent calibration; intervals are not calibrated by the sampler alone. The control supports evaluating this forecasting procedure without assay-derived cell geometry, conditional on H&E-derived proxies and the selected crops. Because both geometry and molecular integration regions change, it neither isolates a causal effect of segmentation nor permits comparing absolute MSE with the primary whole-cell study. It does not validate whole-slide cell detection, biological interpretation of clustering, or optical-spillover correction.

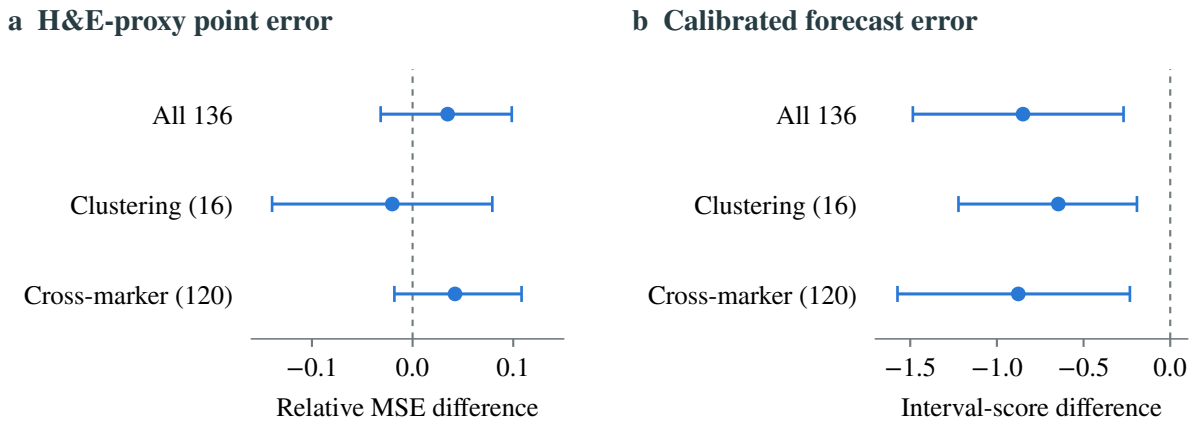


Figure 8: H&E-derived nucleus-plus-rim targets: one seed, five rotations and all 40 patients. Differences are LIGATURE minus the matched direct graph. **a**, Point MSE in this control's own plug-in MSE units; all between-model intervals include zero. **b**, Independently calibrated 90% interval-score differences, with overall coverage 90.6% for both methods. Bars are conditional 95% patient-bootstrap intervals. These targets differ from primary whole-cell targets; this is not whole-slide validation or optical-spillover correction.

Table 24: H&E-derived nucleus-plus-rim targets, one seed, five rotations and all 40 patients. MSE ratios use this control's own plug-in; they are not comparable to ratios or absolute errors from a different target definition. Coverage, training-standard-deviation-normalized width and 90% interval score summarize all 136 statistics. The direct model has lower overall point error; LIGATURE has lower calibrated interval score.

	MSE / own plug-in				
Method / interval	all	same	Coverage	Width	Score
Plug-in / constant	1.000	1.000	90.3%	4.83	7.18
Direct / adaptive	0.925	0.721	90.6%	4.70	6.96
Direct / constant	0.925	0.721	90.3%	4.66	6.94
LIGATURE / adaptive	0.960	0.701	90.6%	4.46	6.11
LIGATURE / constant	0.960	0.701	90.8%	4.70	6.89